HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and tested embedded firmware in **Embedded C** for real-time BMS controllers, implementing low-level drivers for **I²C** and **SPI** communication interfaces.
- Designed and implemented system interfaces using the **CAN** protocol for inter-controller communication within battery management systems.
- Authored **Python** scripts to automate system-level validation tasks and analyze test results, reducing manual testing effort by **~40%**.
- Collaborated with hardware teams to debug and bring up new PCB designs, interpreting **schematics** and component-level hardware designs.
- Supported and validated battery algorithms, including **SoC** and **SoH** estimation, by integrating firmware with simulation environments.
- Interfaced directly with enterprise customers to troubleshoot system integration issues and tune performance parameters for **embedded solutions**.
- Performed lab validation using oscilloscopes and logic analyzers to debug real-time communication issues and ensure adherence to **protocol specifications**.

SKILLS

Embedded Systems – Embedded C, Microcontrollers, Firmware Development, RTOS, Low-Level Drivers, Algorithm Support

Communication Protocols – I²C, SPI, CAN

Programming Languages – Python, Embedded C

Hardware & Tools – PCB Schematics, Oscilloscopes, Function Generators, Logic Analyzers, Debugging Tools

Domain Experience – Battery Management Systems (BMS), Power Electronics, Battery Algorithms (SoC, SoH)

Software Development – System-Level Development, Test Automation, Scripting, Git, CI/CD

PROJECTS

Embedded C Firmware for Real-Time Control

- Developed firmware in **Embedded C** for a micro-controller targeting real-time control applications, focusing on deterministic execution and low latency.
- Implemented interrupt-driven handlers and state machines to manage peripheral interactions, ensuring timely responses to external events.
- Integrated custom drivers for **I²C** and **SPI** peripherals, achieving data throughput of **~1.2 Mbps** for sensor data acquisition.

Python-Based Test Automation for Embedded Systems

- Engineered a **Python** test suite to automate the validation of embedded system firmware, simulating various input conditions and communication patterns.
- Developed scripts to parse log files and analyze test outputs, identifying and reporting defects with **~95% accuracy**.
- Integrated the test suite with a CI/CD pipeline using **Jenkins**, enabling automated regression testing on every code commit.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)